

Other Suggestions

1. A proactive government initiative to encourage buying of home-made products. Domestic electronics product manufacturers must be compelled to use at least 50 per cent of components made in India
2. Use of policy tools like mandatory domestic value-addition and technology transfer in all sectors including consumer products, automotive, and strategic segments like space and defence
3. Special incentive packages that make return on investment attractive as was done with the earlier SIPS scheme, which provided 25 per cent capital subsidy to manufacturing investments meeting certain conditions
4. Development of a strong alliance connecting research labs, academic institutions and the domestic industry
5. Investments, if any, will need to come from the government. The ESDM ecosystem will actually start giving results over a 10-year-plus period. The timescales won't permit venture capitalists, who generally look at two to four years to make significant investments.
6. Special interest subsidy to bring new capacities and help existing manufacturers to expand their capacities
7. Power availability at subsidised rates, with power being a major cost factor in PCB manufacturing

products meeting the protection profile requirements to secure the ICT infrastructure and cyber space of the country

6. Strategic electronics

6.1 Promote manufacturing capacities for sourcing ESDM in strategic sectors—defence, atomic energy, space, etc
(a) Create long-term partnerships between the domestic ESDM industry and strategic sectors for sourcing of electronics products domestically

(b) Defence offset obligations for electronics procurements to be met through ESDM products

7. Create an eco-system for vibrant innovation and R&D in the ESDM sector including nanoelectronics

7.1 Create an electronics development fund to promote innovation and IP and R&D, commercialisation of products, etc in the ESDM, nanoelectronics and IT sectors by providing appropriate

ate funding/incentives to the industry, academia and R&D institutions

(a) Facilitate IP development by the Indian industry, academia and R&D institutions

(b) Set up a 'fund of funds' to create need-based 'daughter funds' for various innovation and manufacturing stages. All funds to be professionally managed

(c) Give special thrust to innovation and R&D for green technologies, convergence and broadband technologies

(d) Promote entrepreneurship in the ESDM sector in active collaboration with the industry, industry associations and academia by ensuring the availability of angel funds and venture capital funds

(e) Set up VLSI-specific incubation centres in four different cities in the country in association with Software Technology Parks of India (STPI), academic institutions or the industry

(f) Design and develop Indian microprocessor for diverse specific and strategic applications

8. Electronics in other sectors

8.1 *Automotive electronics.* Develop a centre of excellence for the development of microcontroller units, micro-electromechanical systems (MEMS) and other advanced electronics devices to enable India to consolidate its position as one of the global auto hubs

8.2 *Avionics.* Support growth of the aviation industry by facilitating the development of R&D and outsourcing of engineering design and related software for avionics and maintenance, repair and overhauling of avionics in the country

8.3 *LED.* Encourage the use of LED lighting solutions, especially in rural markets through innovative products like solar LED lamps, public places like street lighting, traffic lights, etc to promote manufacture of LEDs and LED lights

8.4 *Industrial electronics.* Develop a centre of excellence for innovation in industrial electronics with focus on making affordable standardised products to help India maintain its growth in industrial segments in which it has core competence, including textiles, food processing, steel, engineering and electrical goods like motors, compressors and inverters

8.5 *Medical electronics.* Consolidate the design and development of affordable medical electronics device industry and develop downstream manufacturing activities through sector-specific cluster

8.6 *Solar photovoltaics.* Build manufacturing capacity of solar photovoltaics to support the generation of 20 GW of solar power by 2020

8.7 *Information and broadcasting.* Create an eco-system for manufacture of set-top boxes and other broadcast equipment in the country as part of the digitalisation of the broadcast network of the country

9. Handling e-waste

9.1 Facilitate environment-friendly e-waste handling policies

(a) Create a mechanism with the industry to streamline the implementation of e-waste (Management and Handling) Rules, 2011 including restrictions on usage of hazardous substances as per global best practices

(b) Help streamline procedures to prevent e-waste dumping in the country

(c) Facilitate implementation of Extended Producers Responsibility under the e-waste (Management and Handling) Rules, 2011 for electronics hardware manufacturers as well as recyclers

(d) Promote development of e-waste recycling industry for domestically produced e-waste

(e) Create a specific thrust within Electronic Development Fund for the development of IPR and electronics products in green technologies

10. Governance structures

10.1 Set up a national electronics mission with industry participation to evolve programmes in pursuit of the laid down policies and also to create institutional mechanisms to advance the implementation of various programmes aimed at achieving the objectives enumerated in this policy and promote India as an electronics hardware manufacturing hub and suitably market 'Brand India' in electronics

10.2 Rename the Department of Information Technology as Department of Electronics and Information Technology (DeiTY). ●